

4. Filtering variants for quality

We now have genotypes!

But many of these may not be reliable to to poor coverage
other will be artifacts of thing like poor alignment

#CHROM	POS	Ind1	Ind2	Ind3	Ind4	Ind5	Ind6
Contig1	100	0/0	0/0	0/1	0/1	1/1	1/1
Contig1	200	0/0	0/1	0/1	0/0	0/1	1/1
Contig1	300	1/1	0/1	0/0	0/0	0/1	1/1
Contig1	400	0/0	0/0	0/0	0/1	0/1	0/1
Contig1	500	0/1	0/1	1/1	0/0	0/1	1/1
Contig1	600	0/0	0/1	0/0	1/1	0/1	1/1
Contig1	700	1/1	1/1	0/1	0/1	0/0	0/0
Contig1	800	0/0	0/0	0/1	0/1	0/1	1/1
Contig1	900	0/1	1/1	0/1	0/0	0/0	0/1
Contig1	1000	0/0	0/1	1/1	0/1	0/0	1/1

Two ways we can apply filters

Hard Filtering: The removal of entire *sites* from a vcf file

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Contig1	300	1/1	0/1	0/0	0/0	0/1	1/1
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Contig1	300	1/1	0/1	0/0	0/0	0/1	1/1
Contig1	400	0/0	0/0	0/0	0/1	0/1	0/1
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Common hard filters: **Max depth** (INFO/DP)

If we sequence to expected depth of x , we expect some variation across sites by chance.

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- Collapsed repetitive sequences: several similar genomic regions are represented by one location in the reference, so reads from all copies map there.
- Duplicated genes or paralogues: reads from related gene copies incorrectly map to the same reference position.
- Copy-number variation: some individuals genuinely carry additional copies of the region.

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A Phred-scaled confidence that the site contains a genuine variant.

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Integrates information about

- **Bases observed:** how many reads support the reference and alternative alleles.
- **Base quality:** whether individual bases are likely to have been sequenced correctly.
- **Mapping quality:** whether the reads are likely to have been aligned correctly
- **Read depth:** more reads supporting an allele generally provide stronger evidence.
- **Alignment uncertainty:** bcftools mpileup uses BAQ to reduce the influence of bases that may be misaligned, particularly around indels.
- **Genotype likelihoods:** for each individual, it calculates the likelihood of genotypes such as 0/0, 0/1 and 1/1.
- **Evidence across individuals:** bcftools call combines the genotype likelihoods across samples and applies its population-genetic model to assess whether an alternative allele is present.

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Naively, we expect the forward and reverse strands to be sequenced an equal number of times, but bias can be produced:

- **PCR bias:** some fragments may amplify more successfully in one orientation.
- **Alignment artefacts:** reads from one strand may map differently
- **Read-position effects:** the alternative allele may occur near the ends of reads from one orientation, where error rates are higher.
- **Adapter contamination or incomplete trimming:** residual adapter sequence can affect one read orientation more strongly.
- **Library-preparation biases:** fragmentation, amplification or target-capture steps can produce unequal strand representation.

Soft filters: GQ& DP

Here we are evaluating each genotype based on its individual quality score. For a large VCF this can mean billions of evaluations.

$GQ < 30 \mid DP < 5$

Sample	Original genotype
A	0/1
B	0/1
C	1/1
D	0/0

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Sample	Original genotype	GQ	DP	Result
A	0/1	35	12	0/1
B	0/1	12	10	./.
C	1/1	40	3	./.
D	0/0	30	5	0/0

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After this step its important to remove sites with > some % missing data